Deep

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AI/ML Engineer with hands-on experience building AI-powered applications, GenAI/ML/NLP pipelines, and backend systems using Python. Worked on projects and industry applications involving LLMs, RAG, semantic search, machine learning, MLOps, REST APIs, authentication and authorization, and data analytics using FastAPI, LangChain, HuggingFace, PostgreSQL, and Docker.

EXPERIENCE

Software Engineer · Mphasis Limited *Nov 2026 – Present*
Previously: Trainee Software Engineer (Intern) [GitHub] *Feb 2026 – May 2026*

- Built a full-stack AI-powered ATS using Python, FastAPI, React, and PostgreSQL, automating resume parsing, hybrid candidate scoring, interview scheduling, and end-to-end hiring workflows.
- Engineered a hybrid AI scoring engine combining Semantic Similarity, BM25 TF-IDF, skill-taxonomy matching (80+ skills), and experience relevance scoring to generate explainable candidate rankings — reduced manual screening time by ∼60% (est.).
- Implemented enterprise features including Google OAuth2, JWT authentication, recruiter analytics dashboard, JD optimiser, automated email notifications, and immutable audit logging across a normalised PostgreSQL schema.

Machine Learning Intern · Cognifyz Technology [GitHub] *Jan 2025 – Feb 2025*

- Built restaurant-rating prediction models using Linear Regression and Decision Tree, achieving $R^2 = 0.98$ and reducing RMSE by 35% through feature engineering, target encoding, and normalisation; tracked experiments in MLflow for reproducibility.
- Developed Jaccard-based content recommendation and multi-class cuisine classification models using Decision Tree, improving Precision@5 by ∼28%, handling categorical features and class imbalance, and exposing recom. inference via REST API.

PROJECTS

MediBot — Production RAG Medical Chatbot [GitHub] [Demo]

- Reduced hallucinations on a 2,000+ page medical encyclopaedia by building a production RAG pipeline — PDF extraction, 500-char recursive chunking, 384-dim HuggingFace embeddings (`all-MiniLM-L6-v2`), and Pinecone cosine-similarity search (top-k=3) — with snippet-level source citations (`file, page`) on every response.
- Productionised with Flask + Gunicorn (2 workers), per-IP rate limiting (20 req/min, 200/day), input validation, structured logging, and Docker; sustained 100+ concurrent sessions reliably.
- Improved response reliability by implementing guardrails and system-prompt controls in LangChain to restrict out-of-scope queries; integrated GPT-4o mini for context-aware response generation.

NLP Text Summarisation with MLOps [GitHub]

- Achieved ROUGE-2 of 0.42 by fine-tuning a Pegasus Transformer for abstractive summarisation with automated data ingestion; implemented DVC for data/model versioning and MLflow for experiment tracking, cutting turnaround by ∼50%.
- Exposed the fine-tuned model via FastAPI with Pydantic validation and did real-time inference.

Drishti — AI-Powered Bilingual Assistant for the Visually Impaired *(Capstone)*

- Integrated Whisper ASR, GPT-4o mini, OpenCV, and Flask on a Raspberry Pi to deliver a real-time bilingual (Hindi + English) voice assistant with sub-2 s end-to-end latency; built session-based memory and context-aware follow-up handling.
- Replicated ∼80% of commercial smart-glass features (avg Rs. 2.5L market cost) at a prototype cost of Rs. 7,000 — demonstrating cost-constrained hardware-software system design.

TECHNICAL SKILLS

Languages: C++, Python, SQL

AI / ML & GenAI: Scikit-learn, Pandas, NumPy, Transformers, HuggingFace, LangChain, RAG, Prompt Engineering, Semantic Search, Vector Embeddings, Whisper, GPT-4o mini, Pinecone, MLflow, DVC

Backend & Databases: FastAPI, Flask, REST APIs, JWT Authentication, Google OAuth2, PostgreSQL, MySQL, AsyncIO, Pydantic, Gunicorn

Data Engineering & Analytics: MS SQL Server, Power BI, Power Query, DAX, Star Schema, ETL Pipelines, Data Warehousing, Data Modelling

DevOps & Tools: Docker, Git, GitHub Actions, CI/CD, Linux, OpenCV, Jupyter Notebook

CS Fundamentals: DSA, OOP, DBMS, Software Engineering, Agile/Scrum

Certifications: Kaggle — Intro to Machine Learning (2025) | Tata — GenAI Powered Data Analytics (Forage) | Deloitte Australia — Data Analytics (Forage)

EDUCATION

Course	College / University	Year	CGPA/%
MCA	Thapar Institute of Engineering & Technology (TIET), Patiala	2024 – 2026	8.8/10
B.Sc.	Kurukshetra University, Kurukshetra	2021 – 2024	64.2%
CBSE (Class XII)	The Century School	2018 – 2019	84.6%
CBSE (Class X)	Pratap Public School	2016 – 2017	88.7%